Verificación Capturas:

Contenedores en ejecución:

```
C:\Users\Izan\Desktop\docker24-4>docker-compose ps
time="2026-04-24T15:52:56+02:00" level=warning msg="C:\\Users\\Izan\\Desktop\\docker24-4\\docker-compose.yml: the attribute `version` is obsolete, i
t will be ignored, please remove it to avoid potential confusion"
NAME            IMAGE          COMMAND                  SERVICE        CREATED          STATUS          PORTS
nginx-proxy     nginx:latest   "/docker-entrypoint.…"   nginx-proxy    11 minutes ago   Up 11 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp
web-server-1    nginx:latest   "/docker-entrypoint.…"   web-server-1   11 minutes ago   Up 11 minutes   80/tcp
web-server-2    nginx:latest   "/docker-entrypoint.…"   web-server-2   11 minutes ago   Up 11 minutes   80/tcp
```

Balanceo de carga:

172.21.0.2:80 → web-server-1

172.21.0.3:80 → web-server-2

```
C:\Users\Izan\Desktop\docker24-4>curl.exe -I http://localhost
HTTP/1.1 200 OK
Server: nginx/1.29.8
Date: Fri, 24 Apr 2026 13:53:23 GMT
Content-Type: text/html
Content-Length: 5277
Connection: keep-alive
Last-Modified: Fri, 24 Apr 2026 13:47:55 GMT
ETag: "69eb748b-149d"
Accept-Ranges: bytes
X-Served-By: 172.21.0.2:80


C:\Users\Izan\Desktop\docker24-4>curl.exe -I http://localhost
HTTP/1.1 200 OK
Server: nginx/1.29.8
Date: Fri, 24 Apr 2026 13:53:26 GMT
Content-Type: text/html
Content-Length: 5277
Connection: keep-alive
Last-Modified: Fri, 24 Apr 2026 13:47:55 GMT
ETag: "69eb748b-149d"
Accept-Ranges: bytes
X-Served-By: 172.21.0.3:80
```

```
"Containers": {
    "66171de14c65491c242650867b097320f377cb5eaaa2dc33b1f50332ecc1e898": {
        "Name": "nginx-proxy",
        "EndpointID": "2875146f1227526faefd2aef36cb9cc727169c071f39eb7bc3e66e114ad103dd",
        "MacAddress": "56:9a:54:ad:f7:94",
        "IPv4Address": "172.21.0.4/16",
        "IPv6Address": ""
    },
    "ed24447378fd4dd8baedc8041e20ecdd3c1362ba19d25863db820731aadcc5c8": {
        "Name": "web-server-1",
        "EndpointID": "2ac57bfd39151696c2371d88c2364da8131624240079feac9b9065f2f7c9d269",
        "MacAddress": "e6:5e:18:1d:7a:3b",
        "IPv4Address": "172.21.0.2/16",
        "IPv6Address": ""
    },
    "eee84911cac6130448469d32cbc6dd3061350ac0cffd18ad0fb0b41ac6cd5d06": {
        "Name": "web-server-2",
        "EndpointID": "51a8aadc03dd6950578f93a4ac1e89e909fe0dde73c30284e55f61f9823c0078",
        "MacAddress": "b6:95:53:74:85:b3",
        "IPv4Address": "172.21.0.3/16",
        "IPv6Address": ""
```


web-server-1 funcionando:

```
C:\Users\Izan\Desktop\docker24-4>docker logs -f web-server-1
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/24 13:41:23 [notice] 1#1: using the "epoll" event method
2026/04/24 13:41:23 [notice] 1#1: nginx/1.29.8
2026/04/24 13:41:23 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/24 13:41:23 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/24 13:41:23 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/24 13:41:23 [notice] 1#1: start worker processes
2026/04/24 13:41:23 [notice] 1#1: start worker process 29
2026/04/24 13:41:23 [notice] 1#1: start worker process 30
2026/04/24 13:41:23 [notice] 1#1: start worker process 31
2026/04/24 13:41:23 [notice] 1#1: start worker process 32
2026/04/24 13:41:23 [notice] 1#1: start worker process 33
2026/04/24 13:41:23 [notice] 1#1: start worker process 34
2026/04/24 13:41:23 [notice] 1#1: start worker process 35
2026/04/24 13:41:23 [notice] 1#1: start worker process 36
2026/04/24 13:41:23 [notice] 1#1: start worker process 37
2026/04/24 13:41:23 [notice] 1#1: start worker process 38
2026/04/24 13:41:23 [notice] 1#1: start worker process 39
2026/04/24 13:41:23 [notice] 1#1: start worker process 40
172.21.0.4 - - [24/Apr/2026:13:41:43 +0000] "GET / HTTP/1.1" 200 2067 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
 Gecko) Chrome/147.0.0.0 Safari/537.36" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:24 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:28 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:29 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:31 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:32 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:46:56 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:48:24 +0000] "GET / HTTP/1.1" 200 5277 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:48:55 +0000] "GET / HTTP/1.1" 200 5277 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
 Gecko) Chrome/147.0.0.0 Safari/537.36" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:48:55 +0000] "GET /imagenes/about_time.jpg HTTP/1.1" 200 136647 "http://localhost/" "Mozilla/5.0 (Windows NT 10.0; Wi
n64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:48:55 +0000] "GET /imagenes/the_hunt.jpg HTTP/1.1" 200 4503 "http://localhost/" "Mozilla/5.0 (Windows NT 10.0; Win64;
```

web-server-2 funcionando:

```
C:\Users\Izan\Desktop\docker24-4>docker logs -f web-server-2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/24 13:41:24 [notice] 1#1: using the "epoll" event method
2026/04/24 13:41:24 [notice] 1#1: nginx/1.29.8
2026/04/24 13:41:24 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/24 13:41:24 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/24 13:41:24 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/24 13:41:24 [notice] 1#1: start worker processes
2026/04/24 13:41:24 [notice] 1#1: start worker process 29
2026/04/24 13:41:24 [notice] 1#1: start worker process 30
2026/04/24 13:41:24 [notice] 1#1: start worker process 31
2026/04/24 13:41:24 [notice] 1#1: start worker process 32
2026/04/24 13:41:24 [notice] 1#1: start worker process 33
2026/04/24 13:41:24 [notice] 1#1: start worker process 34
2026/04/24 13:41:24 [notice] 1#1: start worker process 35
2026/04/24 13:41:24 [notice] 1#1: start worker process 36
2026/04/24 13:41:24 [notice] 1#1: start worker process 37
2026/04/24 13:41:24 [notice] 1#1: start worker process 38
2026/04/24 13:41:24 [notice] 1#1: start worker process 39
2026/04/24 13:41:24 [notice] 1#1: start worker process 40
2026/04/24 13:41:45 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.21.0.4, server:
localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost", referrer: "http://localhost/"
172.21.0.4 - - [24/Apr/2026:13:41:45 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) Apple
WebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:28 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:29 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:31 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:32 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:45:42 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:46:56 +0000] "HEAD / HTTP/1.1" 200 0 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:48:52 +0000] "GET / HTTP/1.1" 200 5277 "-" "curl/8.13.0" "172.22.0.1"
172.21.0.4 - - [24/Apr/2026:13:48:55 +0000] "GET /imagenes/dune2.jpg HTTP/1.1" 200 58515 "http://localhost/" "Mozilla/5.0 (Windows NT 10.0; Win64; x
64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "172.22.0.1"
```

Red aislada:

Línea clave: “Internal”: true, los contenedores que estén conectados con esta flag pueden hablar entre sí pero docker prohibirá el tráfico de salida hacia el exterior.

```
C:\Users\Izan\Desktop\docker24-4>docker network inspect docker24-4_backend_net
[
    {
        "Name": "docker24-4_backend_net",
        "Id": "6d4985bdc8535a81287894506cd72d3032689777405f4dbc48a98f682d4b80a2",
        "Created": "2026-04-24T13:41:22.392488057Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.21.0.0/16",
                    "Gateway": "172.21.0.1"
                }
            ]
        },
        "Internal": true,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
```

Curl directo a los servidores backend, vemos que al intentar conectarnos a cualquiera de los dos nos da un timeout, esto quiere decir que el host no sabe como llegar a la subred del backend.

Por otro lado sabemos que el propio proxy tiene un mapeo explicito donde indica que el mismo es la única puerta abierta al host.

```
PS C:\Users\Izan\Desktop\docker24-4> docker ps --format "table {{.Names}}\t{{.Ports}}"
NAMES          PORTS
nginx-proxy    0.0.0.0:80->80/tcp, [::]:80->80/tcp
web-server-1   80/tcp
web-server-2   80/tcp
proxy-squid    0.0.0.0:3128->3128/tcp, [::]:3128->3128/tcp
juego_db       3306/tcp
PS C:\Users\Izan\Desktop\docker24-4> $IP1 = docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}'
 web-server-1
>> echo "Intentando conectar directamente a web-server-1 ($IP1)..."
>> curl.exe --connect-timeout 2 http://$IP1
Intentando conectar directamente a web-server-1 (172.21.0.2)...
curl: (28) Connection timed out after 2004 milliseconds
PS C:\Users\Izan\Desktop\docker24-4> $IP1 = docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}'
 web-server-2
>> echo "Intentando conectar directamente a web-server-1 ($IP1)..."
>> curl.exe --connect-timeout 2 http://$IP1
Intentando conectar directamente a web-server-1 (172.21.0.3)...
curl: (28) Connection timed out after 2005 milliseconds
```

Proxy funcionando:

Al web-server-1:

```
C:\Users\Izan\Desktop\docker24-4>docker exec -it nginx-proxy curl http://172.21.0.2
<!DOCTYPE html>
<html lang="es">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Infraestructura Docker - Galería Multimedia</title>
    <style>
        :root {
            --primary: #4facfe;
            --secondary: #00f2fe;
            --bg: #0f172a;
            --card-bg: rgba(30, 41, 59, 0.7);
        }
```

Al web-server-2:

```
C:\Users\Izan\Desktop\docker24-4>docker exec -it nginx-proxy curl http://172.21.0.3
<!DOCTYPE html>
<html lang="es">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Infraestructura Docker - Galería Multimedia</title>
    <style>
        :root {
            --primary: #4facfe;
            --secondary: #00f2fe;
            --bg: #0f172a;
            --card-bg: rgba(30, 41, 59, 0.7);
        }
```